

INDUSTRIAL DIESEL GENERATOR SET

FIELD SERVICE and MAINTENANCE MANUAL

Document Reference: MNL-GEN-2026-G650

Equipment Class: Diesel Engine-Driven Standby Generator

Version: 1.0.0

SECTION 1: SYSTEM OVERVIEW

The diesel generator set produces electrical power by coupling a diesel engine to an alternator. The major components a field technician services are: the diesel engine, the alternator (generator end), the fuel system, the cooling system (radiator and coolant), the starting system (battery and starter), the lubrication system, and the control panel with the automatic voltage regulator (AVR).

Safety: Shut down and lock out the set before servicing. Exhaust gases are lethal in enclosed spaces; ensure ventilation. Fuel is flammable; the cooling system is hot and pressurized; the battery can deliver high current and produces explosive gas.

SECTION 2: DIESEL ENGINE

Component identity: The diesel engine is the prime mover that drives the alternator.

Failure mode 2.1 — Engine cranks but will not start

Symptoms: The starter turns the engine but it does not fire.

Cause: No fuel, air in the fuel system, clogged fuel filter, or a fuel solenoid fault.

Corrective steps:

1. Confirm there is adequate fuel in the tank and the fuel valve is open.
2. Replace a clogged fuel filter and bleed air from the fuel system.
3. Verify the fuel shut-off solenoid energizes (opens) on a start command.
4. Check for water in the fuel/water separator and drain it.

Failure mode 2.2 — Engine overheating

Symptoms: High coolant temperature alarm or shutdown; steam from the radiator.

Cause: Low coolant, blocked radiator, failed water pump, or a slipping/broken belt.

Corrective steps:

1. Shut down and allow the engine to cool before opening the cooling system.
2. Check the coolant level and top up to the specified mark.

3. Clean debris from the radiator core and confirm airflow.
4. Inspect the water-pump drive belt tension and condition (see Section 4).

Failure mode 2.3 — Low oil pressure shutdown

Symptoms: The set shuts down on a low-oil-pressure alarm.

Cause: Low oil level, a clogged oil filter, or a failing oil pump.

Corrective steps:

1. Shut down and check the engine oil level on the dipstick; top up to the mark.
2. Replace the oil filter if it is overdue or clogged.
3. If oil level and filter are correct but pressure stays low, the oil pump or sender requires inspection.

SECTION 3: ALTERNATOR (GENERATOR END)

Component identity: The alternator converts the engine's mechanical rotation into electrical output, regulated by the AVR.

Failure mode 3.1 — Engine runs but no electrical output

Symptoms: The engine runs normally but there is no voltage at the output terminals.

Cause: Failed AVR, loss of residual magnetism, or a winding/connection fault.

Corrective steps:

1. Shut down and lock out. Check the output circuit breaker is closed and terminals are tight.
2. Inspect the AVR and its connections; a failed AVR is a common cause of no output.
3. If equipped, follow the procedure to re-flash residual magnetism.
4. Test the alternator windings if output is still absent after the AVR is verified.

Failure mode 3.2 — Output voltage too high, too low, or unstable

Symptoms: Output voltage out of range or fluctuating under load.

Cause: AVR misadjustment or fault, or unstable engine speed (governor).

Corrective steps:

1. Verify engine speed/frequency is stable at the rated value (a speed problem looks like a voltage problem).
2. Check and adjust the AVR voltage setting to the specified output.
3. Replace the AVR if voltage remains unstable with correct engine speed.

SECTION 4: COOLING SYSTEM

Component identity: The cooling system (radiator, coolant, water pump, fan, and belt) removes engine heat.

Failure mode 4.1 — Coolant leak or low coolant

Symptoms: Coolant pooling under the set; low coolant level; overheating.

Cause: Failed hose, loose clamp, leaking water pump seal, or radiator damage.

Corrective steps:

1. Shut down and let the system cool and depressurize before opening.
2. Locate the leak; tighten clamps or replace a failed hose.
3. Replace the water pump if it leaks at the weep hole.
4. Refill with the specified coolant mix and check the level after a run.

Failure mode 4.2 — Loose or broken drive belt

Symptoms: Squealing, overheating, or no water-pump/fan rotation.

Cause: Belt wear or incorrect tension.

Corrective steps:

1. Shut down and lock out. Inspect the belt for cracks, glazing, or breakage.
2. Replace a worn or broken belt.
3. Set belt tension to the specified deflection.

SECTION 5: STARTING SYSTEM (BATTERY AND STARTER)

Component identity: The starting system uses a battery and starter motor to crank the engine.

Failure mode 5.1 — Engine will not crank

Symptoms: Nothing or only a click when starting; slow cranking.

Cause: Discharged or failed battery, corroded terminals, or a failed starter.

Corrective steps:

1. Check the battery voltage and state of charge; charge or replace as needed.
2. Clean and tighten the battery terminals and the starter cable connections.
3. If the battery is good and connections are clean but the engine will not crank, test or replace the starter motor.

SECTION 6: LUBRICATION SYSTEM

Component identity: The lubrication system circulates engine oil to reduce wear.

Failure mode 6.1 — Oil leak or high oil consumption

Symptoms: Oil under the engine; falling oil level between services.

Cause: Failed gasket or seal, loose drain plug, or a leaking filter.

Corrective steps:

1. Shut down. Clean the engine to locate the leak source.
2. Tighten the drain plug and re-seat or replace the oil filter if leaking there.
3. Replace a failed gasket or crankshaft seal if oil leaks from a joint or seal.

SECTION 7: CONTROL PANEL

Component identity: The control panel manages starting, monitoring, protection shutdowns, and the AVR.

Failure mode 7.1 — Spurious shutdown or no start from the panel

Symptoms: The set shuts down without an obvious fault, or will not start from the panel.

Cause: A tripped protection sensor, a wiring fault, or a control fuse.

Corrective steps:

1. Read the fault/shutdown code displayed on the control panel.
2. Address the indicated fault (for example low oil pressure or high temperature per the relevant section).
3. Check control-panel fuses and sensor wiring; reset the controller and retest.

SECTION 8: ROUTINE MAINTENANCE SCHEDULE

Daily / before run: Check engine oil and coolant levels; check fuel level; drain the fuel/water separator; verify the battery is charged.

Weekly: Inspect for leaks (oil, coolant, fuel); check the drive belt; clean the radiator core.

Monthly: Test-run the set under load; check battery terminals; verify the control panel alarms function.

Quarterly: Change engine oil and filters per hours run; inspect the alternator and AVR connections; sample coolant condition.

